LIANA AKOBIAN

Département d'études cognitives, École Normale Supérieure, 29 Rue d'Ulm, 75005 Paris
liana.akobian@ens.psl.eu

EDUCATION AND TRAINING

École Normale Supérieure, Paris, France — Oct 2026 - present
PhD, Computational Neuroscience
Advisor: N. Alex Cayco-Gajic
Funded by an EDSC doctoral fellowship

Vienna University of Technology, Vienna, Austria — Feb 2021 - Feb 2025
Master of Science, Data Science
Thesis: Modeling Dynamics on a Canonical Neural Manifold in *C. elegans*.
Advisors: Manuel Zimmer, Peter Filzmoser

Katholieke Universiteit Leuven, Leuven, Belgium — Feb - Jul 2023
Erasmus+ Exchange, Artifical Intelligence

University of Vienna, Vienna, Austria — Oct 2016 - Jan 2021
Bachelor of Science, Computer Science

RESEARCH EXPERIENCE

École Normale Supérieure, Paris, France. — Sep 2025 - Sep 2026
Research Associate, Cayco-Gajic Lab, Département d'études cognitives

Applied and validated SchurMO, a method using Riemannian gradient descent for finding interpretable Schur forms within connectivity matrices of recurrent neural networks.
Presented this work at Cosyne 2026 and co-authored a paper accepted at NeurIPS (Oral).

University of Vienna, Vienna, Austria. — Oct 2023 - Nov 2024
Master Thesis, Zimmer Lab, Department of Neurobiology

Aligned multiple heterogeneous calcium imaging recordings of *C. elegans* and generation of neural manifolds using PCA and state-of-the-art methods such as CEBRA and LFADS. Extended and optimized dLDS using control theory and gradient descent.
Authored thesis and presented it at a workshop for statistics.

University of Vienna and **Vienna University of Technology**, Vienna, Austria. — Sep 2022 - Feb 2023
Research Internship, Zimmer Lab, Department of Neurobiology and CSTAT, Statistics and Mathematical Methods in Economics

Analysed and explored calcium imaging data of *C. elegans*, using functional data analysis methods, such as FPLSR and FGLM, and machine learning tools like SVM and decision trees.

GRANTS & AWARDS

2026–2029 PhD Fellowship (Contrat doctoral), Doctoral School of Cognitive Sciences (ED 662), Université PSL. Fully funded for 3 years and awarded through a competitive selection.

PUBLICATIONS

Alexander Lanine, **Liana Akobian**, N Alex Cayco Gajic. *Learning interpretable Schur forms of recurrent weight matrices.* NeurIPS 2026 (Oral).

PRESENTATIONS

Alexander Lanine, **Liana Akobian**, N Alex Cayco Gajic. Poster: *SchurMO: recovering functionally feed-forward structure in recurrent weights through manifold optimization* Bernstein, Germany, 2026. (presenting author)

Alexander Lanine, **Liana Akobian**, N Alex Cayco Gajic. Poster: *Learning interpretable Schur forms of recurrent weight matrices.* Cosyne, Portugal, 2026. (58% acceptance rate) (co-presenting author)

Liana Akobian. Contributed Talk: *Uncovering Neural Control: A Dynamical Systems Approach to Disentangling Intrinsic and Controlled Neural Dynamics.* Workshop on Neural Dynamical Systems and Time-Series Data, QUARIMAFI @ U.Wien, WPI, Inst CNRS Pauli, Austria, 2025.

SCIENTIFIC ORGANISATION

Co-Organiser, Google Developer Group on Campus TU Vienna. Dec 2024 - Jul 2025
Curation and hosting of scientific and industrial talks related to Machine Learning and LLMs.

OTHER WORK EXPERIENCE

Data Engineer, Vienna University of Technology. Jan 2021 - Jan 2023
Management of a Big Data cluster for academic and industrial researchers. Presentation of courses and workshops on big data handling and processing. Automation of cluster configurations using orchestration tools like Ansible.

Working Student, SAP. Oct 2018 - Aug 2020
Management of innovative technical projects such as Chatbots based on knowledge-based systems for HR.

Intern, Verbund. Sep 2018
Development of web applications (Javascript, PL/SQL) and handling of data security and privacy.

Intern, Böhringer Ingelheim. Jul - Aug 2018
Development of a web service (Java, Javascript, SQL) used to plan and organise lab management tasks and orders.

SKILLS AND INTERESTS

Programming	Python, Java, R, C++, Julia, Javascript, SQL, HTML, Visual Basic
Machine Learning	PyTorch, Tensorflow, Keras, Scitkit-Learn, Numpy, Pandas, WandB, JAX
Languages	German (native), Armenian (native), English (native-level proficiency), French (beginner)